CONCRETE:

1. UNLESS OTHERWISE NOTED IN THE SCHEDULES OR DETAILS, THE MINIMUM 28 DAY COMPRESSIVE STRENGTH OF CONCRETE SHALL BE 4000 PSI FOR ALL PARTS OF THE STRUCTURE EXCEPT THE FOOTINGS AND PILE CAPS SHALL BE 3000 PSI. ALL CONCRETE SHALL BE NORMAL WEIGHT (150 PCF) EXCEPT FOR COMPOSITE SLABS WHICH SHALL BE LIGHTWEIGHT CONCRETE WITH A MAXIMUM WEIGHT OF 115 PCF. 3
2. ALL CONCRETE EXPOSED TO THE WEATHER SHALL BE AIR-ENTRAINED. ALL OTHER CONCRETE MAY BE AIR-ENTRAINED OR NON-AIR-ENTRAINED AT THE CONTRACTORS OPTION. FOR SURFACE FINISHES AND OTHER REQUIREMENTS, REFER TO THE SPECIFICATIONS.
3. DETAILS OF FABRICATION OF REINFORCEMENT, HANDLING AND PLACEMENT OF THE CONCRETE, CONSTRUCTION OF FORMS AND PLACEMENT OF REINFORCEMENT, NOT OTHERWISE COVERED BY THE PLANS AND SPECIFICATIONS, SHALL COMPLY WITH THE A.C.I. CODE REQUIREMENTS.
4. HORIZONTAL CONSTRUCTION JOINTS IN THE CONCRETE WALLS OTHER THAN THOSE SHOWN ON THE PLANS WILL NOT BE PERMITTED WITHOUT WRITTEN PERMISSION FROM THE RESIDENT ENGINEER. ADEQUATE PROVISION SHALL BE MADE FOR SECURING FORMS BELOW JOINTS TO PREVENT LEAKAGE AND BULGING OF THE CONCRETE.
5. NO HORIZONTAL CONSTRUCTION JOINTS SHALL BE MADE IN THE SLABS OR BEAMS. VERTICAL CONSTRUCTION JOINTS, MADE AS DESCRIBED BELOW, SHALL BE LOCATED NEAR THE MIDDLE OF THE SPANS. ALL SUCH VERTICAL JOINTS SHALL BE KEYED.
6. WHERE A JOINT IS TO BE MADE, THE SURFACE OF THE CONCRETE SHALL BE THOROUGHLY CLEANED AND ALL LATIANCE REMOVED. IN ADDITION THE JOINTS SHALL BE THOROUGHLY WETTED AND SLUSHED WITH A COAT OF NEAT CEMENT GROUT IMMEDIATLY BEFORE PLACING CONCRETE.
7. ALL ITEMS OF WORK TO BE INSTALLED IN ANY CONCRETE WORK, INCLUDING PIPES SLEEVES AND ELECTRICAL CONDUITS, ETC. SHALL BE PROPERLY LOCATED, INSTALLED AND CHECKED BEFORE PLACING CONCRETE.
8. PROVIDE 3/4" CHAMFERS ON ALL EXPOSED EDGES OF CONCRETE AND THE EXPOSED CORNERS OF BEAMS, GIRDERS AND COLUMNS UNLESS OTHERWISE SHOWN OR NOTED. SEE ARCHITECTURAL DRAWINGS FOR ADDITIONAL DETAILS.
9. THE CONTRACTOR SHALL CONSULT WITH THE RESIDENT ENGINEER BEFORE STARTING CONCRETE WORK TO ESTABLISH A SATISFACTORY PLACING SCHEDULE AND TO DETERMINE THE LOCATION OF CONSTRUCTION JOINTS SO AS TO MINIMIZE THE EFFECTS OF SHRINKAGE IN THE FLOOR SLAB.
10. DO NOT BACKFILL AGAINST CONCRETE WALLS UNTIL THE SUPPORTING TOP SLAB AND BOTTOM SLAB AND THE WALL, HAVE ATTAINED AT LEAST 2/3 OF THE DESIGNED COMPRESSIVE STRENGTH.
11. ALL SIZES AND LOCATIONS OF SLAB OPENINGS AND CURBS FOR MECHANICAL EQUIPMENT SHALL BE VERIFIED WITH THE CONTRACTOR.
12. REFER TO THE ARCHITECTURAL DRAWINGS FOR THE EXACT EXTENT OF ALL DOVETAIL ANCHOR SLOTS, DRIP INSERTS, AND LOCATIONS FOR ALL REGLETS REQUIRED IN THE CONCRETE.
13. ALL HOLES DRILLED INTO THE CONCRETE FOR DOWELS SHALL BE TREATED AS FOLLOWS:
    A. DRILL HOLES 1/4" LARGER THAN BAR OR BOLT TO BE EMBEDDED.
    B. DRILL HOLES WITH SINGLE CHISEL TOOTH ROTARY PERCUSSION DRILL THAT FEEDS COMPRESSED AIR TO THE BASE OF THE HOLE THOUGH A HOLLOW STEM DRILL BIT.
    C. DRILL THE HOLE A MINIMUM OF 15 BAR DIAMETERS DEEP OR AS SHOWN ON THE PLANS.
    D. BLOW OUT THE HOLE WITH COMPRESSED AIR TO REMOVE ALL DUST AND LOOSE MATERIALS
    E. CONSULT WITH A/E FOR FURTHER RECOMMENDATIONS.

FOUNDATIONS:

1. AUGER CAST PILES ARE DESIGNED FOR THE FOLLOWING ALLOWABLE LOADS:

ALLOWABLE DOWNWARD VERTICAL LOAD (TONS PER PILE)			
PILE TIP ELEV.	16" DIA.	18" DIA.	
670.0	25	30	PILE LENGTH SHALL NOT BE LESS THAN 20'-0"
660.0	50	60	
650.0	75	90	
640.0	100	120	

ALLOWABLE UPWARD VERTICAL LOAD (TONS PER PILE)			
PILE TIP ELEV.	16" DIA.	18" DIA.	
660.0	15	17	
650.0	27.5	31	
640.0	40	45	

REINFORCING STEEL

1. REINFORCEMENT, OTHER THAN WELDED WIRE FABRIC, SHALL BE A.S.T.M. A615, GRADE 60.
2. THE CONTRACTOR SHALL FIGURE TO PROVIDE, FABRICATE AND PLACE 15.0 TONS OF REINFORCING STEEL IN ADDITION TO THAT SHOWN ON THE PLANS. THIS STEEL IS TO BE USED AT THE DIRECTION OF THE RESIDENT ENGINEER. THE OWNER SHALL BE GIVEN CREDIT AT THE CONTRACT UNIT PRICE FOR THE UNUSED PORTION. PROVIDE THE OWNER AND A/E WITH THE VALUE OF THE UNIT PRICE.
3. PROVIDE 2-#5 BARS AT THE JAMBS AND 2-#7 BARS AT THE THE HEAD OF ALL OPENINGS IN CONCRETE WALLS UNLESS OTHERWISE SHOWN OR NOTED ON THE PLANS. THESE BARS SHALL EXTEND 2'-0" BEYOND THE OPENING LIMITS OR BE PROVIDED WITH 90 DEGREE HOOKS.
4. THE SHOP DRAWINGS FOR REINFORCING STEEL SHALL INCLUDE SCALE ELEVATIONS OF ALL CONCRETE WALLS.
5. WHERE WALLS SIT ON FOUNDATIONS PROVIDE DOWELS FOR THE WALL. FOR SIZE AND SPACING OF DOWELS SEE SECTIONS AND DETAILS. PROVIDE CORNER BARS AT ALL WALL CORNERS AND WALL INTERSECTIONS.
6. AT ALL OPENINGS IN SOLID SLABS PROVIDE 2-#6 X OPENING WIDTH + 4'-0" IN THE MIDDLE OF THE SLAB AT ALL SIDES OF THE OPENING UNLESS OTHERWISE SHOWN OR NOTED ON THE PLANS.
7. WHERE MAIN SLAB REINFORCEMENT IS INTERRUPTED BY OPENINGS, PLACE THE EQUIVALENT OF ONE HALF OF THE INTERRUPTED STEEL IN FULL LENGTH BARS ON EACH SIDE OF THE OPENING UNLESS OTHERWISE SHOWN.
8. WHERE THE PRINCIPAL REINFORCEMENT IN FLOOR OR ROOF SLABS EXTENDS IN ONE DIRECTION ONLY, TEMPERATURE REINFORCEMENT SHALL BE PROVIDED. SUCH REINFORCEMENT SHALL BE EQUIVALENT TO .0018 TIMES THE CONCRETE CROSS SECTIONAL AREA.
9. WHERE HOOKS ARE INDICATED, PROVIDE STANDARD HOOKS PER A.C.I. AND C.R.S.I. FOR ALL BARS UNLESS OTHER HOOK DIMENSIONS ARE SHOWN ON THE PLANS OR DETAILS.
10. MINIMUM CONCRETE COVER OVER REINFORCING STEEL SHALL BE AS FOLLOWS: 3" AT FOUNDATIONS, 2" AT ALL DIRT FACES OF WALLS, 1-1/2" AT ALL PIER TIES AND WALLS EXPOSED TO THE WEATHER AND 3/4" AT OTHER WALL FACES AND IN STRUCTURAL SLABS UNLESS SHOWN OR NOTED OTHERWISE.
11. WELDED WIRE FABRIC SHALL MEET ALL REQUIREMENTS OF ASTM A185.2. ALL WELDED WIRE FABRIC SHALL BE FURNISHED IN A MAT FORM ONLY.

LINTELS

1. WHERE LINTELS ARE NOT SPECIFICALLY SHOWN OR NOTED ON THE STRUCTURAL OR ARCHITECTURAL DRAWINGS, PROVIDE THE FOLLOWING LINTELS OVER ALL OPENINGS AND RECESS IN BOTH INTERIOR AND EXTERIOR WALLS.
   A) BRICK: FOR OPENINGS UP TO 8'-0" PROVIDE ONE 6" X 4" X 3/8" ANGLE FOR EACH 4" THICKNESS OF WALL. FOR THESE LINTELS PROVIDE 1" BEARING FOR EACH FOOT OF SPAN WITH A MINIMUM OF 8" BEARING. FOR OPENINGS OVER 8'-0" SEE THE DRAWINGS.
   B) BLOCK: WHERE BLOCK WILL NOT BE EXPOSED TO VIEW IN THE FINISHED ROOM USE STANDARD PRECAST CONCRETE LINTELS WITH A MINIMUM OF 8" BEARING. WHERE THE BLOCK IS EXPOSED IN THE FINISHED ROOM, USE LINTEL BLOCK FILLED WITH CONCRETE. GROUT ALL EXPOSED JOINTS AND REINFORCE AS FOLLOWS:
      1) FOR 4" THICK BLOCK: 2-#3 BARS FOR OPENINGS TO 5'-0"
      2) FOR 6" THICK BLOCK: 2-#3 BARS FOR OPENINGS TO 4'-8"
      3) FOR 8" THICK BLOCK: 2-#4 BARS FOR OPENINGS TO 5'-0"
2. CONSULT A/E FOR REINFORCING IN LARGER OPENINGS OR IF WALL IS TO BE A BEARING WALL.

DESIGN:

1. THE FOLLOWING GRAVITY LOADS WHERE USED IN THE DESIGN OF THE BUILDING:

   LIVE LOADS:
   A) PUBLIC AREAS:....................100 PSF
   B) PATIENT AREAS:....................100 PSF
   C) MECHANICAL ROOMS:.................150 PSF
   D) INTERSTITIAL ARES:.................25 PSF
   E) ROOFS:.............. 30 PSF + SNOW DRIFT
   F) ROOF OVER AREA 'B' AS NOTED ON THE PLANS.
   G) BULK STORAGE ROOMS 150PSF
   ADDITIONAL DEAD LOAD ALLOWANCES:
   A) CEILINGS:...........................5 PSF
   B) SUSPENDED M/E LOADS:..............12 PSF
   C) PARTITIONS........................20 PSF
   D) ACTUAL WEIGHT OF MECHANICAL EQUIPMENT ON ROOF AS NOTED ON THE PLANS.
2. THIS BUILDING HAS BEEN DESIGNED AS AN ESSENTIAL FACILITY, EARTHQUAKE ZONE 1, PER THE 1988 INDIANA BUILDING CODE AND V.A. HANDBOOK H-08-8.
3. THIS BUILDING HAS BEEN DESIGNED USING A 75 MPH WIND LOAD PER THE 1988 INDIANA BUILDING CODE.

PROVISIONS FOR FUTURE EXPANSION:

COLUMNS AND FOUNDATIONS ARE DESIGNED FOR FUTURE VERTICAL EXPANSION AS FOLLOWS:

AREA D: ENTIRE AREA NORTH OF LINE 21 IS DESIGNED FOR SIX ADDITIONAL FLOORS PLUS ROOF SLAB.

AREA E: ENTIRE AREA BETWEEN LINES A AND F.3 IS DESIGNED FOR TWO ADDITIONAL FLOORS PLUS ROOF SLAB.

AREA B: AREA SOUTH OF LINE 14 AND STAIR NO. 10 IS DESIGNED TO EXTEND UP TO THE EIGHTH FLOOR PLUS ROOF SLAB.

AREA F: ENTIRE AREA ABOVE SIXTH FLOOR IS DESIGNED FOR TWO ADDITIONAL FLOORS PLUS ROOF SLAB.

STRUCTURAL STEEL

1. ALL STRUCTURAL STEEL TO BE A.S.T.M. A36. ALL TUBE SECTIONS SHALL BE A.S.T.M. A500, GRADE B.
2. DETAILS FOR DESIGN, FABRICATION AND ERECTION OF ALL STRUCTURAL STEEL SHALL BE IN ACCORDANCE WITH THE A.I.S.C. STANDARDS UNLESS OTHERWISE NOTED OR SPECIFIED.
3. PROVIDE TEMPORARY ERECTION BRACING AS REQUIRED.
4. ALL SHOP CONNECTIONS SHALL BE WELDED. SEE SPECIFICATIONS FOR BOLT AND WELD REQUIREMENTS. FIELD CONNECTIONS SHALL BE ACCOMPLISHED USING HIGH STRENGTH A-325 BOLTS UNLESS OTHERWISE NOTED. PROVIDE ERECTION BOLTS AS REQUIRED FOR WELDED CONNECTIONS.
5. UNLESS OTHERWISE SHOWN OR NOTED, HIGH STRENGTH BOLTS SHALL BE 3/4" DIAMETER.
6. PERMANENT MACHINE BOLTS, USING AN APPROVED TYPE OF SELF ANCHORING HEX NUT, MAY BE USED FOR SUCH MINOR CONNECTIONS AS SHELF ANGLES, CLOSURES, ETC.
7. UNLESS OTHERWISE SHOWN OR NOTED ON THE PLANS, PROVIDE 8" BEARING AT EACH END FOR ALL BEAMS.
8. FOR LOOSE LINTELS, MASONRY SHELF ANGLES AND OTHER SUCH ITEMS GENERALLY NOT SHOWN ON THE STRUCTURAL PLANS, SEE THE ARCHITECTURAL PLANS. SEE GENERAL NOTES THIS SHEET FOR LINTEL SIZE AND REINFORCING.
9. ~~MILL THE BEARING SURFACES OF ALL COMPRESSION MEMBERS.~~
10. ALL STRUCTURAL STEEL SHALL BE FIRE-PROOFED UNLESS LOCATED IN MASONRY WALLS. REFER TO SPECIFICATIONS. 2

METAL ROOF DECK

1. ALL ROOF DECK IS TO BE 1-1/2", 20 GAGE, S.D.I. STANDARD, TYPE B, GALVANIZED METAL DECK WITH THE FOLLOWING REQUIREMENTS: (WIDE RIB)
   MIN. I = 0.210 in4
   MIN. SP= 0.232 in3
   MIN. SN= 0.245 in3
2. DECK SHALL BE PROVIDED IN A MINIMUM OF 3 SPAN LENGTHS WHERE POSSIBLE.
3. DECK IS DESIGNED AS A HORIZONTAL DIAPHRAGM. ALL WELDS TO BE 5/8" PUDDLE WELDS. WELDING PATTERN SHALL BE AS FOLLOWS:
   A. AT SHEET ENDS---------12" O.C.
   B. AT EDGE OF DECK------24" O.C.
   C. AT INTERIOR SUPPORT--12" O.C.
4. PROVIDE #12 CADMIUM PLATED SHEET METAL SCREWS AT 24" O.C. FOR ALL SIDE LAPS.
5. DECK TO CONFORM TO ALL SDI CRITERIA AND RECOMENDATIONS. SEE SPECIFICATIONS FOR ADDITIONAL INFORMATION.
6. PROVIDE SUPPORT FOR ALL EDGES OF DECK WHETHER SHOWN ON THE DRAWINGS OR NOT.

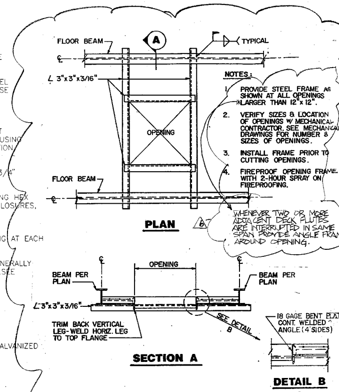
COMPOSITE METAL DECK

1. FLOOR SLAB CONSTRUCTION TO BE 5-1/4" LIGHTWEIGHT CONCRETE SLABS WITH 6" X 6", W2.1 X W2.1 W.W.F. ON 2"X 20 GAGE (MINIMUM) GALVANIZED COMPOSITE METAL DECK, UNLESS NOTED OTHERWISE ON PLANS OR DETAILS.
   A) 20 GAGE DECK TO MEET THE FOLLOWING REQUIREMENTS:
      6" WIDE TROUGHS AT 12" o.c.
      I = 0.43 in4
      SP = 0.369 in3
      SN = 0.38 in3
2. SCREEDS SHALL BE SET TAKING INTO ACCOUNT THE DEAD LOAD DEFLECTIONS NOTED ON THE PLANS AND ANY CAMBER IN THE BEAMS IN ORDER TO ACHIEVE A LEVEL FLOOR. SET BEAMS AND GIRDERS WITH CAMBER UP.
3. THE SEQUENCE AND METHOD OF POURING THE FLOORS SHALL BE SUBMITTED TO THE RESIDENT ENGINEER PRIOR TO STARTING ANY CONCRETE OPERATIONS. IN GENERAL, CONCRETE POURS SHOULD BE MADE ACROSS THE ENTIRE WIDTH OF A FLOOR IN A DIRECTION PERPENDICULAR TO THE GIRDERS. ANY SINGLE POUR SHOULD BE STOPPED BETWEEN BEAMS (NEAREST TO COLUMN).
4. SHEAR CONNECTORS SHALL BE 3/4" DIA. X 3-7/8" LONG HEADED STUDS. SEE PLANS FOR QUANTITIES.
5. CONTRACTOR SHALL SUBMIT SHOP DRAWINGS INDICATING SHEAR STUD LOCATIONS ON EACH BEAM.

EQUIPMENT PADS

PROVIDE 4" CONCRETE PAD WITH 6x6-W1.4xW1.4 WELDED WIRE FABRIC UNDER MECHANICAL EQUIPMENT AS SHOWN ON MECHANICAL DRAWINGS. SEE MECHANICAL DRAWINGS FOR PLAN SIZE AND LOCATIONS.

HANGER ATTACHMENT TO THE INTERSTITIAL DECK SCREW SHALL BE A #10-16 HWH SDS/2 3/4". THE PLATE THAT THE SCREW AND HANGER WIRE IS ATTACHED TO SHALL BE AS USED IN THE FIELD TEST, HOWEVER, PLATE SHALL BE FABRICATED FROM FLAT STOCK AND NOT FIELD BENT. UNDER NO CIRCUMSTANCES SHALL ANY HANGER LOAD EXCEED 50 POUNDS. 4



TYPICAL FRAMING @ INTERSTITIAL FLOOR OPENING 1

8-1-2-1296

Revisions	Date
1 AMENDMENT #2, ITEM 152 A	7/25/91
2 AMENDMENT #2, ITEM 126 A	7/25/91
3 AMENDMENT #3, ITEM A	7/25/91
4 VA# -S-114	10/21/92
6 VA# - M-198	11/11/92
6 CLARIFICATION	9/18/92
RECORD DRAWING	

BOYD|SOBIERAY ASSOCIATES, inc.
indianapolis, indiana
BSA architects engineers health planners

DALE L. JACOBS REGISTERED No. 12562 STATE OF INDIANA PROFESSIONAL ENGINEER

Drawing Title	Project Title	Date
GENERAL NOTES	CLINICAL IMPROVEMENTS (WTS)/ MENTAL HEALTH & BEHAVIORAL SCIENCES CENTER (CSR) PHASE IV	3/18/91
Approved Project Director	Building Number: 1 / Checked: RDP / Drawn: RAB	583-042
	Location: INDIANAPOLIS,INDIANA WTS	DRAWING NO. 1-S01 Dwg. 1 Of 65

Office of Facilities
Department of Veterans Affairs

VA FORM 08-6231A, JAN 1990

PLOT DATE: 12/19/90